

RRC-Raptor-V4H

- SDK Toolchain

V0.0.2

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i. VERSION LIST

Description	Version	Date	Author
SDK Toolchain	V0.0.1	2024/08/30	kevinlin
File attribute changes	V0.0.2	2025/01/22	kevinlin

ii. HASHTAG

#V4H

#BSP

#YOCTO

Toolchain

1. Overview

Target :

Mainly for the RRC-Raptor project using Kernel 5.10.147 environment under the YOCTO 3.1.11 version to build the SDK and compile the environment.

2. System Requirements

PC software:

Distributor : Ubuntu
Description : Ubuntu 20.04.4 LTS
Release : 20.04
Codename : focal
Kernel : 5.15.0-43-generic

PC hardware:

PC : Intel i9-11900K @ 2.50GHz
Width : 64 bits
Memory : 64G
Block device : 7.3TB Disk

3. SDK Toolchain build setup

Information :

Use YOCTO i.MX Yocto-5.15.71-2.0.0 version ◦

Download and Build environment :

```
$ mkdir imx-yocto-bsp
$ cd imx-yocto-bsp
$ repo init -u https://github.com/nxp-imx/imx-manifest -b
imx-linux-kirkstone -m imx-5.15.71-2.0.0.xml
$ repo sync

$ DISTRO=fsl-imx-xwayland MACHINE=imx6dlsabresd source
imx-setup-release.sh -b build
```

3.1. SDK Toolchain Building

Step :

In the YOCTO compilation directory.

Specify toolchain recipes to execute compilation instructions.

```
$ bitbake meta-toolchain
```

After completion, the file will be built in <build_directory>/tmp/deploy/sdk/
poky-glibc-x86_64-meta-toolchain-core2-64-qemux86-64-toolchain-3.1.11.sh

toolchain Version :

glibc 3.1.11

gcc core2-64

3.1.1. SDK Toolchain Information

Execution information :

```
$ ./poky-glibc-x86_64-meta-toolchain-core2-64-qemux86-64-toolchain-3.1.11.sh -h
```

Usage: {FILE}.sh [-y] [-d <dir>]

-y Automatic yes to all prompts

-d <dir> Install the SDK to <dir>

===== Extensible SDK only options =====

-n Do not prepare the build system

-p Publish mode (implies -n)

===== Advanced DEBUGGING ONLY OPTIONS =====

-S Save relocation scripts

-R Do not relocate executables

-D use set -x to see what is going on

-l list files that will be extracted

3.2. SDK Toolchain

Install :

Go to the working environment platform to be executed and install the toolchain under /opt/poky/3.1.11. You can use the following commands to set some environment variables.

```
$ ./poky-glibc-x86_64-meta-toolchain-core2-64-qemux86-64-toolchain-3.1.11.sh  
-d /opt/poky/3.1.11
```

The toolchain required for the entire compilation will be installed.

After completion, you can see that there are some files placed under /opt/poky/3.1.11, some are environment variable configuration files, and the toolchain is all in the sysroot folder.

3.3. Compile

example :

If the SDK is installed in /opt/poky/3.1.11

We need to setup environment in a particular shell. You have to do it once for every new shell window you open :

```
$ . /opt/poky/3.1.11/environment-setup-core2-64-poky-linux
```

Compile hello_world.c :

```
$ $CC hello_world.c -o hello_world
```

Check executable file info :

```
$ file hello_world
```

4. APPENDIX